

1. General Description

The EM74HC259; EM74HCT259 is an 8-bit addressable latch. The device features four modes of operation. In the addressable latch mode, data on the D input is written into the latch addressed by the inputs A0 to A3. The addressed latch will follow the data input, non-addressed latches will retain their previous states. In memory mode, all latches retain their previous states and are unaffected by the data or address inputs. In the 3-to-8 decoding or demultiplexing mode, the addressed output follows the D input and all other outputs are LOW. In the reset mode, all outputs are forced LOW and unaffected by the data or address inputs. Inputs include clamp diodes. This enables the use of current limiting resistors to interface inputs to voltages in excess of V_{CC} .

2. Features and Benefits

- Wide supply voltage range from 2.0 V to 6.0 V
- Latch-up performance exceeds 250 mA
- Complies with JEDEC standards:
 - JESD8C (2.7 V to 3.6 V)
 - JESD7A (2.0 V to 6.0 V)
- Combined demultiplexer and 8-bit latch
- Serial-to-parallel capability
- Output from each storage bit available
- Random (addressable) data entry
- Easily expandable
- Common reset input
- Useful as a 3-to-8 active HIGH decoder
- Input levels:
 - For EM74HC259: CMOS level
 - For EM74HCT259: TTL level

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Product datasheet, Rev. 1.0

Apr 28, 2025

■ ESD protection:

- HBM: ANSI/ESDA/JEDEC JS-001 class 2 exceeds 3500 V
- CDM: ANSI/ESDA/JEDEC JS-002 class C3 exceeds 2000 V

■ Multiple package options

■ Specified from -40 °C to +85 °C and from -40 °C to +125 °C

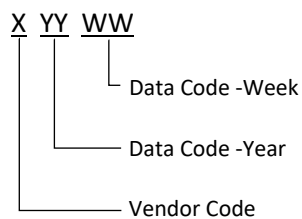
3. Ordering Information

Table 1. Ordering information

Type number	Topside marking	Package		
		Name	Description	Quantity
EM74HC259D	HC259 XYWW	SOP-16L	plastic small outline package; 16 leads; body width 3.9 mm	3000
EM74HCT259D	HCT259 XYWW			
EM74HC259PW	HC259 XYWW	TSSOP-16L	plastic thin shrink small outline package; 16 leads; body width 4.4 mm	3000
EM74HCT259PW	HCT259 XYWW			

MARKING INFORMATION

NOTE: XYWW = Vendor Code and Data Code.



4. Function Diagram

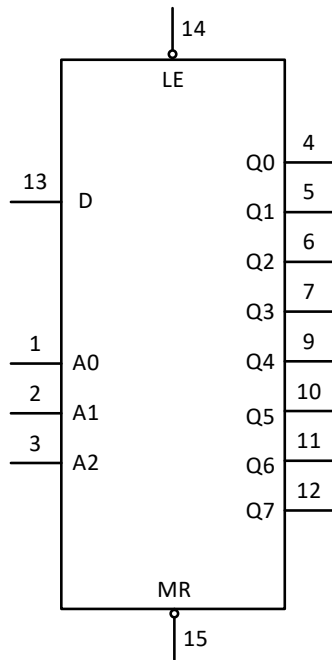


Fig. 1. Logic symbol

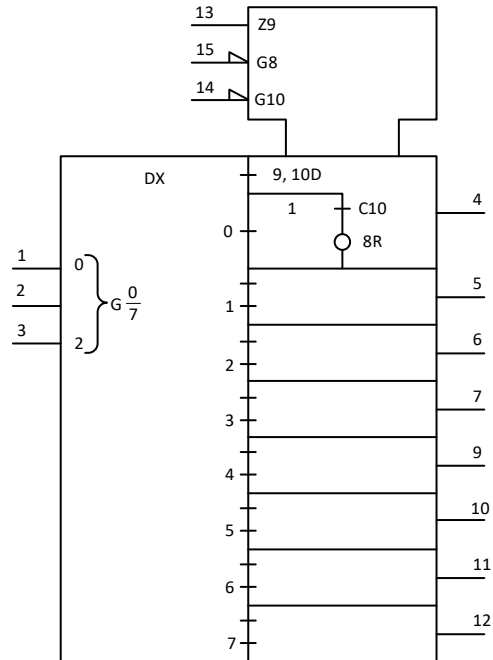


Fig. 2. IEC logic symbol

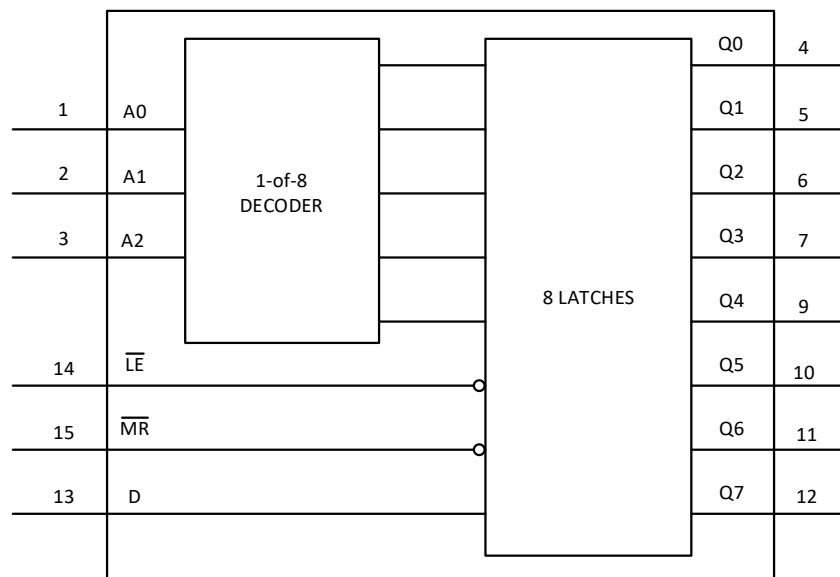
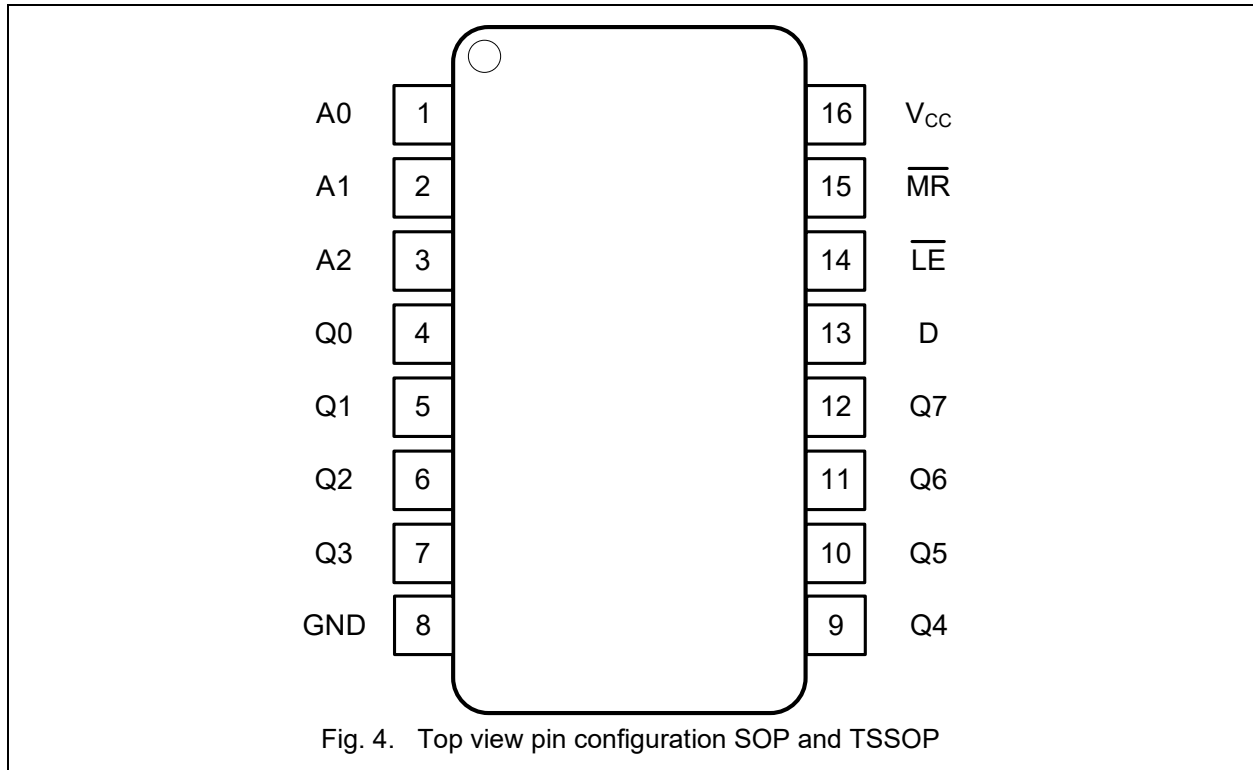


Fig. 3. Functional diagram

5. Pinning Information

5.1. Pinning



5.2. Pin description

Table 2. Pin description

Symbol	Pin	Description
A0, A1, A2	1, 2, 3	Address input
Q0, Q1, Q2, Q3, Q4, Q5, Q6, Q7	4, 5, 6, 7, 9, 10, 11, 12	Latch output
GND	8	Ground (0 V)
D	13	Data input
$\overline{LE}$	14	Latch enable input (active LOW)
$\overline{MR}$	15	Conditional reset input (active LOW)
V _{CC}	16	Supply voltage

6. Functional Description

Table 3. Function table

H = HIGH voltage level; L = LOW voltage level; X = don't care;

d = HIGH or LOW data one set-up time prior to the LOW-to-HIGH LE transition;

q = lower case letter indicates the state of the referenced input one set-up time prior to the LOW-to-HIGH transition.

Operating mode	Input						Output							
	$\overline{\text{MR}}$	$\overline{\text{LE}}$	D	A0	A1	A2	Q0	Q1	Q2	Q3	Q4	Q5	Q6	Q7
Reset (clear)	L	H	X	X	X	X	L	L	L	L	L	L	L	L
Demultiplexer (active HIGH 8-channel) decoder (when D = H)	L	L	d	L	L	L	Q=d	L	L	L	L	L	L	L
	L	L	d	H	L	L	L	Q=d	L	L	L	L	L	L
	L	L	d	L	H	L	L	L	Q=d	L	L	L	L	L
	L	L	d	H	H	L	L	L	L	Q=d	L	L	L	L
	L	L	d	L	L	H	L	L	L	L	Q=d	L	L	L
	L	L	d	H	L	H	L	L	L	L	L	Q=d	L	L
	L	L	d	L	H	H	L	L	L	L	L	L	Q=d	L
Memory (no action)	H	H	X	X	X	X	q ₀	q ₁	q ₂	q ₃	q ₄	q ₅	q ₆	q ₇
Addressable latch	H	L	d	L	L	L	Q=d	q ₁	q ₂	q ₃	q ₄	q ₅	q ₆	q ₇
	H	L	d	H	L	L	q ₀	Q=d	q ₂	q ₃	q ₄	q ₅	q ₆	q ₇
	H	L	d	L	H	L	q ₀	q ₁	Q=d	q ₃	q ₄	q ₅	q ₆	q ₇
	H	L	d	H	H	L	q ₀	q ₁	q ₂	Q=d	q ₄	q ₅	q ₆	q ₇
	H	L	d	L	L	H	q ₀	q ₁	q ₂	q ₃	Q=d	q ₅	q ₆	q ₇
	H	L	d	H	L	H	q ₀	q ₁	q ₂	q ₃	q ₄	Q=d	q ₆	q ₇
	H	L	d	L	H	H	q ₀	q ₁	q ₂	q ₃	q ₄	q ₅	Q=d	q ₇
	H	L	d	H	H	H	q ₀	q ₁	q ₂	q ₃	q ₄	q ₅	q ₆	Q=d

Table 4. Operating mode select table

H = HIGH voltage level; L = LOW voltage level.

$\overline{\text{LE}}$	$\overline{\text{MR}}$	Mode
L	H	Addressable latch mode
H	H	Memory mode
L	L	Demultiplexer mode
H	L	Reset mode

7. Absolute Maximum Ratings

Stresses exceeding the absolute maximum ratings may damage the device. The device may not function or be operable above the recommended operating conditions and stressing the parts to these levels is not recommended. In addition, extended exposure to stresses above the recommended operating conditions may affect device reliability. The absolute maximum ratings are stress ratings only.

Table 4. Absolute Maximum Ratings

In accordance with the Absolute Maximum Rating System (IEC 60134). Voltages are referenced to GND.

Symbol	Parameter	Conditions	Min	Max	Unit
V_{CC}	supply voltage		-0.5	7	V
I_{IK}	input clamping current	$V_I < -0.5\text{ V}$ or $V_I > V_{CC} + 0.5\text{ V}$		± 20	mA
I_{OK}	output clamping current	$V_O < -0.5\text{ V}$ or $V_O > V_{CC} + 0.5\text{ V}$		± 20	mA
I_O	output current	$V_O = -0.5\text{ V}$ to $(V_{CC} + 0.5\text{ V})$		± 25	mA
I_{CC}	supply current			70	mA
I_{GND}	ground current		-70		mA
P_{tot}	total power dissipation	$T_{amb} = -40\text{ }^{\circ}\text{C}$ to $+125\text{ }^{\circ}\text{C}$		500	mW
T_{stg}	storage temperature		-65	150	$^{\circ}\text{C}$

[1] The input and output voltage ratings may be exceeded if the input and output current ratings are observed.

8. Recommended Operating Conditions

The Recommended Operating Conditions table defines the conditions for actual device operation. Recommended operating conditions are specified to ensure optimal performance to the datasheet specifications. EnergyMath does not recommend exceeding them or designing to Absolute Maximum Ratings.

Table 5. Recommended Operating Conditions

Symbol	Parameter	Conditions	EM74HC259			EM74HCT259			Unit
			Min	Typ	Max	Min	Typ	Max	
V_{CC}	supply voltage		2.0	5.0	6.0	4.5	5.0	5.5	V
V_I	input voltage		0		V_{CC}	0		V_{CC}	V
V_O	output voltage		0		V_{CC}	0		V_{CC}	V
T_{amb}	ambient temperature		-40	25	125	-40	25	125	$^{\circ}\text{C}$
$\Delta t/\Delta V$	Input transition rise and fall rate	$V_{CC} = 2.0\text{ V}$			625				ns/V
		$V_{CC} = 4.5\text{ V}$		1.67	139		1.67	139	ns/V
		$V_{CC} = 6.0\text{ V}$			83				ns/V

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9.Static Characteristics

Table 6. Static characteristics

At recommended operating conditions. Voltages are referenced to GND (ground = 0 V).

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HC259								
V _{IH}	HIGH-level input voltage	V _{CC} = 2.0 V	1.5			1.5		V
		V _{CC} = 4.5 V	3.15			3.15		V
		V _{CC} = 6.0 V	4.2			4.2		V
V _{IL}	LOW-level input voltage	V _{CC} = 2.0 V			0.5		0.5	V
		V _{CC} = 4.5 V			1.35		1.35	V
		V _{CC} = 6.0 V			1.8		1.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = -20 μA; V _{CC} = 2.0 V	1.9			1.9		V
		I _O = -20 μA; V _{CC} = 4.5 V	4.4			4.4		V
		I _O = -20 μA; V _{CC} = 6.0 V	5.9			5.9		V
		I _O = -4.0 mA; V _{CC} = 4.5 V	3.84			3.7		V
		I _O = -5.2 mA; V _{CC} = 6.0 V	5.34			5.2		V
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL}						
		I _O = 20 μA; V _{CC} = 2.0 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 4.5 V			0.1		0.1	V
		I _O = 20 μA; V _{CC} = 6.0 V			0.1		0.1	V
		I _O = 4.0 mA; V _{CC} = 4.5 V			0.33		0.4	V
		I _O = 5.2 mA; V _{CC} = 6.0 V			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 6.0 V			±1		±1	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 6.0 V			20		40	μA
C _I	input capacitance			6.5				pF

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
EM74HCT259								
V _{IH}	HIGH-level input voltage	V _{CC} = 4.5 V to 5.5 V	2.0			2.0		V
V _{IL}	LOW-level input voltage	V _{CC} = 4.5 V to 5.5 V			0.8		0.8	V
V _{OH}	HIGH-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = -20 μA	4.4		4.4		V	
		I _O = -4.0 mA	3.84		3.7		V	
V _{OL}	LOW-level output voltage	V _I = V _{IH} or V _{IL} ; V _{CC} = 4.5 V						
		I _O = 20 μA			0.1		0.1	V
		I _O = 4.0 mA			0.33		0.4	V
I _I	input leakage current	V _I = V _{CC} or GND ; V _{CC} = 5.5 V			±1		±1	μA
I _{CC}	supply current	V _I = V _{CC} or GND ; I _O = 0 A ; V _{CC} = 5.5 V			20		40	μA
ΔI _{CC}	additional supply current	V _I = V _{CC} - 2.1 V; other inputs at V _{CC} or GND; V _{CC} = 4.5 V to 5.5 V; I _O = 0 A			450		490	μA
C _I	input capacitance			6.5				pF

10. Dynamic Characteristics

Table 7. Dynamic characteristics

Voltages are referenced to GND (ground = 0 V); for test circuit see Fig. 11.

Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
EM74HC259								
t _{pd}	propagation delay	D to Qn; see Fig. 5 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		20	ns
		V _{CC} = 6.0 V			13		15	ns
		An to Qn; see Fig. 6 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		20	ns
		V _{CC} = 6.0 V			13		15	ns
		$\overline{\text{LE}}$ to Qn; see Fig. 7 [2]						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		20	ns
		V _{CC} = 6.0 V			13		15	ns
t _{PHL}	HIGH to LOW propagation delay	$\overline{\text{MR}}$ to Qn; see Fig. 8						
		V _{CC} = 2.0 V			55		60	ns
		V _{CC} = 4.5 V			17		20	ns
		V _{CC} = 6.0 V			13		15	ns
t _t	transition time	see Fig. 7 [3]						
		V _{CC} = 2.0 V			8		10	ns
		V _{CC} = 4.5 V			7		9	ns
		V _{CC} = 6.0 V			7		9	ns

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ[1]	Max	Min	Max	
t _w	pulse width	$\overline{LE}$ HIGH or LOW; see Fig. 7						
		V _{CC} = 2.0 V	100			120		ns
		V _{CC} = 4.5 V	20			24		ns
		V _{CC} = 6.0 V	17			20		ns
		$\overline{MR}$ LOW; see Fig. 8						
		V _{CC} = 2.0 V	75			90		ns
		V _{CC} = 4.5 V	15			18		ns
		V _{CC} = 6.0 V	13			15		ns
t _{su}	set-up time	D, An to $\overline{LE}$; see Fig. 9 and Fig. 10						
		V _{CC} = 2.0 V	75			90		ns
		V _{CC} = 4.5 V	15			18		ns
		V _{CC} = 6.0 V	13			15		ns
t _h	hold time	D to $\overline{LE}$; see Fig. 9 and Fig. 10						
		V _{CC} = 2.0 V	5			5		ns
		V _{CC} = 4.5 V	5			5		ns
		V _{CC} = 6.0 V	5			5		ns
		An to $\overline{LE}$; see Fig. 9 and Fig. 10						
		V _{CC} = 2.0 V	5			5		ns
		V _{CC} = 4.5 V	5			5		ns
		V _{CC} = 6.0 V	5			5		ns
C _{PD}	power dissipation capacitance	f = 1 MHz; V _I = GND to V _{CC} [4]		26				pF
EM74HCT259								
t _{pd}	propagation delay	D to Q _n ; see Fig. 5 [2]						
		V _{CC} = 4.5 V			17		20	ns
		An to Q _n ; see Fig. 6 [2]						
		V _{CC} = 4.5 V			17		20	ns
		$\overline{LE}$ to Q _n ; see Fig. 7 [2]						
		V _{CC} = 4.5 V			17		20	ns

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Symbol	Parameter	Conditions	-40 °C to +85 °C			-40 °C to +125 °C		Unit
			Min	Typ	Max	Min	Max	
t_{PHL}	HIGH to LOW propagation delay	$\overline{MR}$ to Qn; see Fig. 8						
		$V_{CC} = 4.5$ V			17		20	ns
t_t	transition time	see Fig. 7 [3]						
		$V_{CC} = 4.5$ V			7		9	ns
t_w	pulse width	$\overline{LE}$ HIGH or LOW; see Fig. 7						
		$V_{CC} = 4.5$ V	20			24		ns
		$\overline{MR}$ LOW; see Fig. 8						
		$V_{CC} = 4.5$ V	31			38		ns
t_{su}	set-up time	D, An to $\overline{LE}$; see Fig. 9 and Fig. 10						
		$V_{CC} = 4.5$ V	15			18		ns
t_h	hold time	D to $\overline{LE}$; see Fig. 9 and Fig. 10						
		$V_{CC} = 4.5$ V	5			5		ns
		An to $\overline{LE}$; see Fig. 9 and Fig. 10						
		$V_{CC} = 4.5$ V	5			5		ns
C_{PD}	power dissipation capacitance	$f = 1$ MHz; $V_I = GND$ to $V_{CC} - 1.5$ V [4]		28				pF

[1] Typical values are measured at nominal supply voltage ($V_{CC} = 3.3$ V and $V_{CC} = 5.0$ V).

[2] t_{pd} is the same as t_{PLH} and t_{PHL} .

[3] t_t is the same as t_{THL} and t_{TLH} .

[4] C_{PD} is used to determine the dynamic power dissipation (P_D in μW).

$P_D = C_{PD} \times V_{CC}^2 \times f_i \times N + \Sigma(C_L \times V_{CC}^2 \times f_o)$ where:

f_i = input frequency in MHz;

f_o = output frequency in MHz;

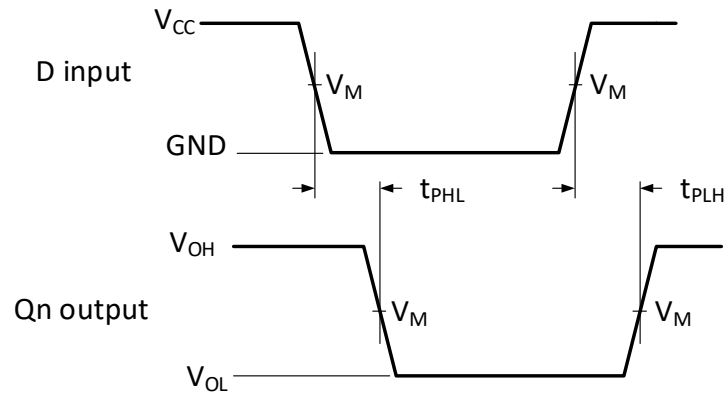
C_L = output load capacitance in pF;

V_{CC} = supply voltage in V;

N = number of inputs switching;

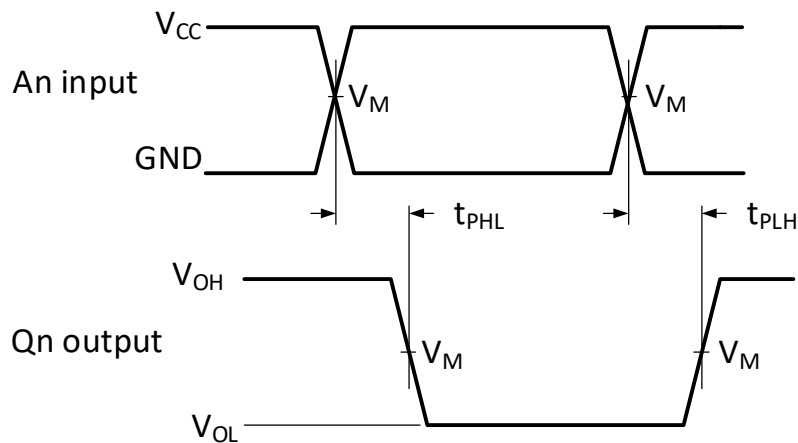
$\Sigma(C_L \times V_{CC}^2 \times f_o)$ = sum of outputs.

10.1. Waveforms and test circuit



Measurement points are given in Table 8.
 V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 5. Data input to output propagation delays

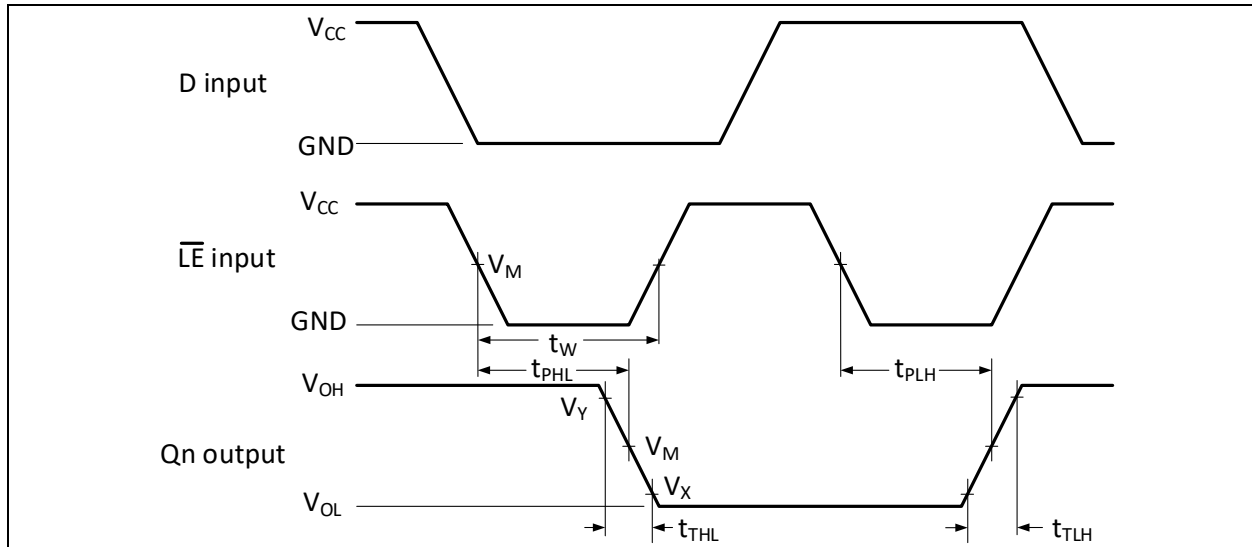


Measurement points are given in Table 8.
 V_{OL} and V_{OH} are typical output voltage levels that occur with the output load.

Fig. 6. Address input to output propagation delays

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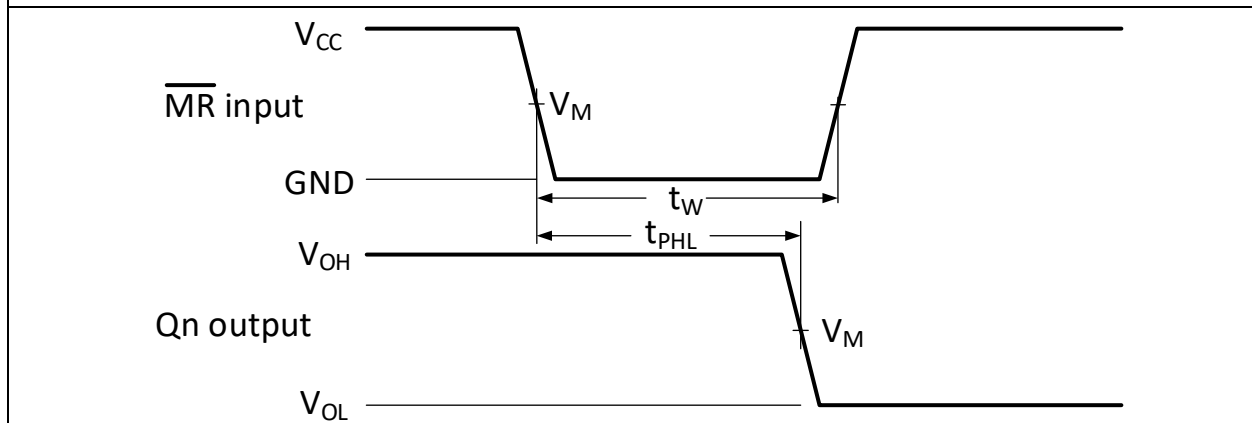
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Measurement points are given in Table 8.

V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

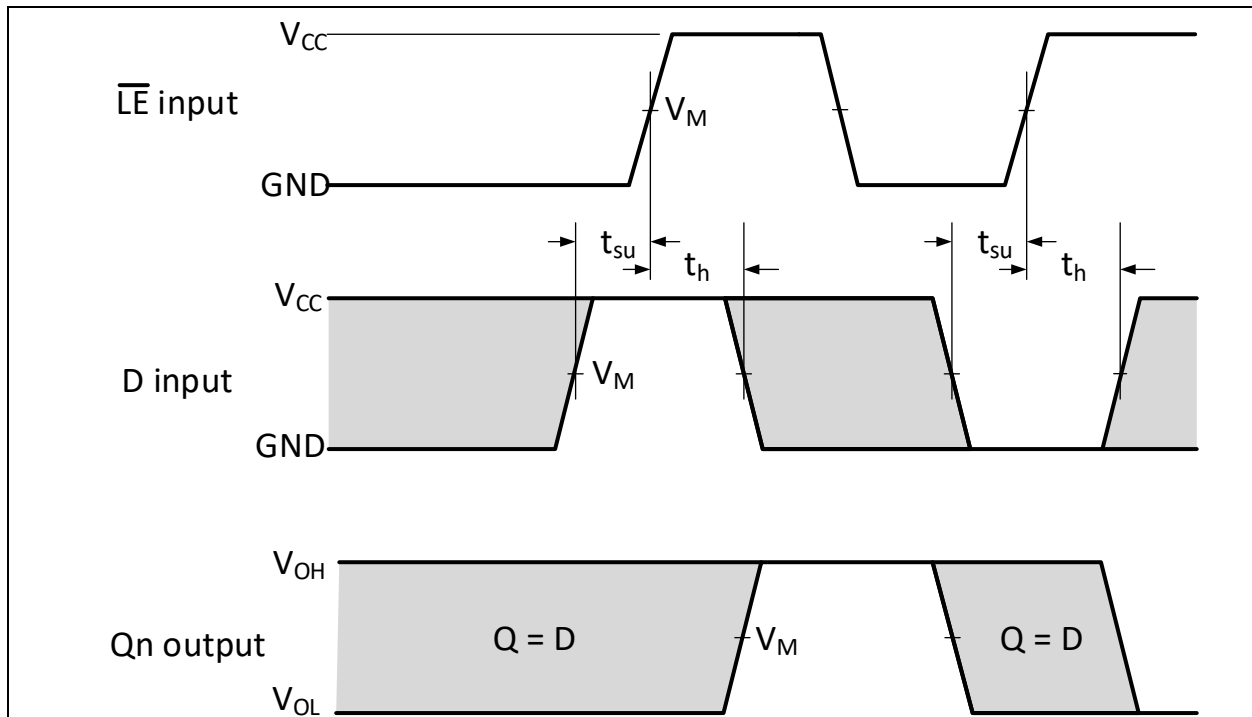
Fig. 7. Enable input to output propagation delays and pulse width



Measurement points are given in Table 8.

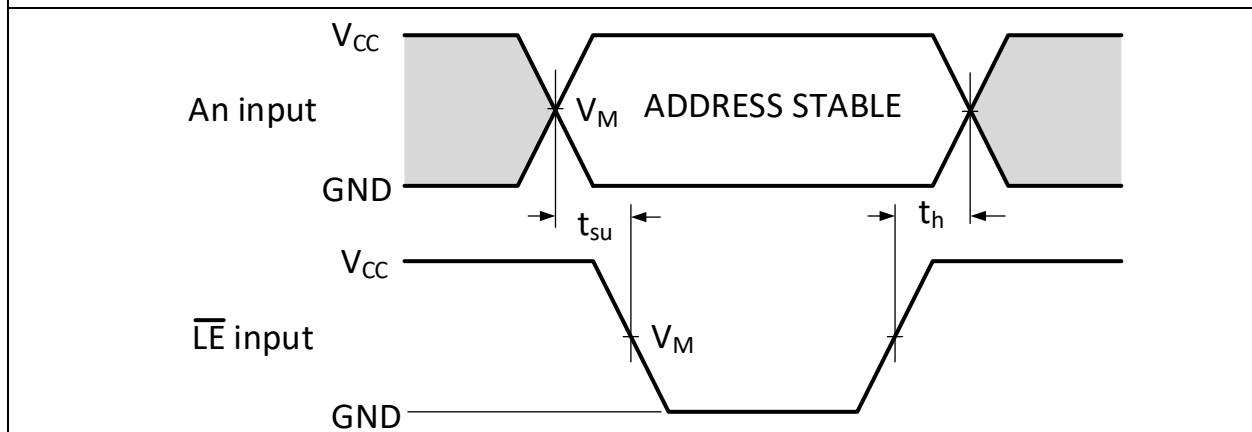
V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

Fig. 8. Master reset input to output propagation delays



Measurement points are given in Table 8.
The shaded areas indicate when the input is permitted to change for predictable output performance.
 V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

Fig. 9. Data input to latch enable input set-up and hold times



Measurement points are given in Table 8.
The shaded areas indicate when the input is permitted to change for predictable output performance.
 V_{OL} and V_{OH} are typical voltage output levels that occur with the output load.

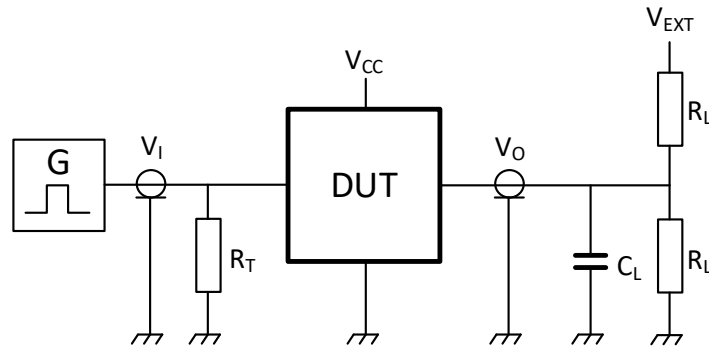
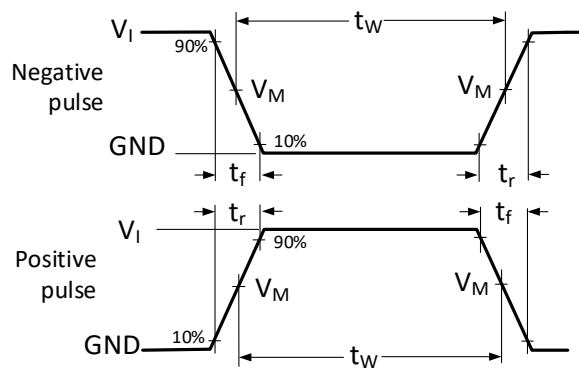
Fig. 10. Address input to latch enable input set-up and hold times

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Table 8. Measurement points

Type	Input	Output		
	V_M	V_M	V_X	V_Y
EM74HC259	$0.5V_{CC}$	$0.5V_{CC}$	$0.1V_{CC}$	$0.9V_{CC}$
EM74HCT259	1.3 V	1.3 V	$0.1V_{CC}$	$0.9V_{CC}$



Test data is given in Table 9.

Definitions for test circuit:

R_L = Load resistance.

C_L = Load capacitance including jig and probe capacitance.

R_T = Termination resistance should be equal to the output impedance Z_o of the pulse generator.

V_{EXT} = External voltage for measuring switching times.

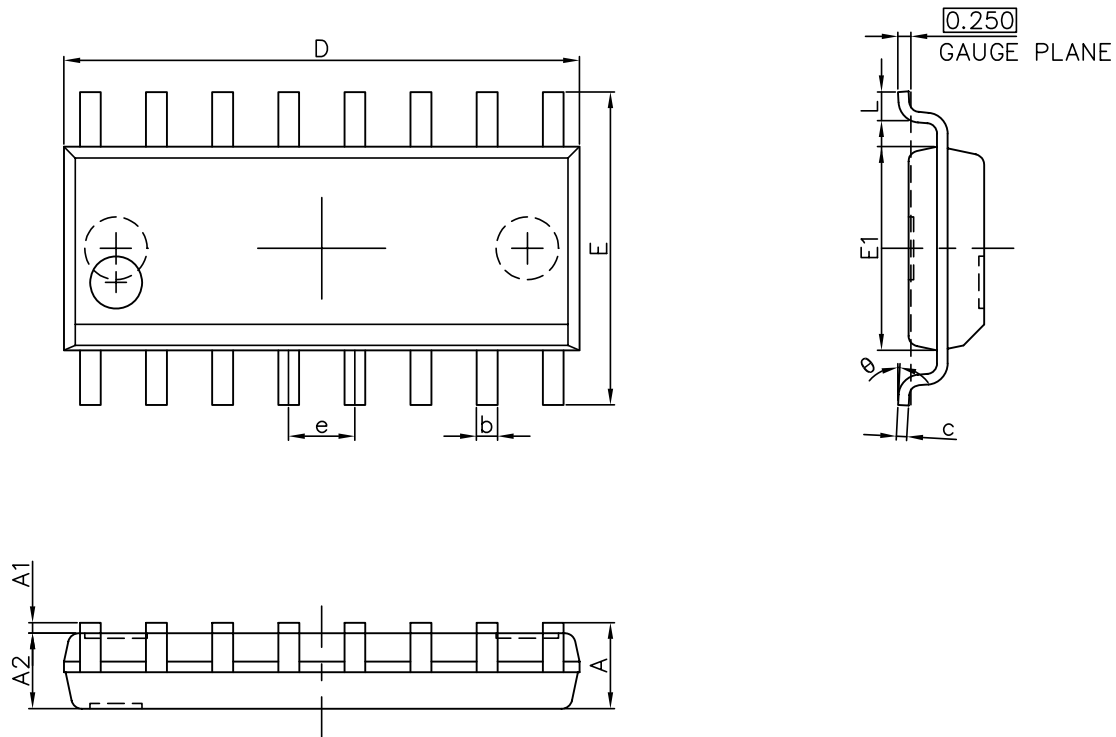
Fig. 11. Test circuit for measuring switching times

Table 9. Test data

Type	Input		Load		V_{EXT}
	V_I	$t_r = t_f$	C_L	R_L	t_{PLH}, t_{PHL}
EM74HC259	V_{CC}	2.0 ns	50 pF	500 Ω	open
EM74HCT259	3 V	2.0 ns	50 pF	500 Ω	open

11. Package Outline

SOP-16L

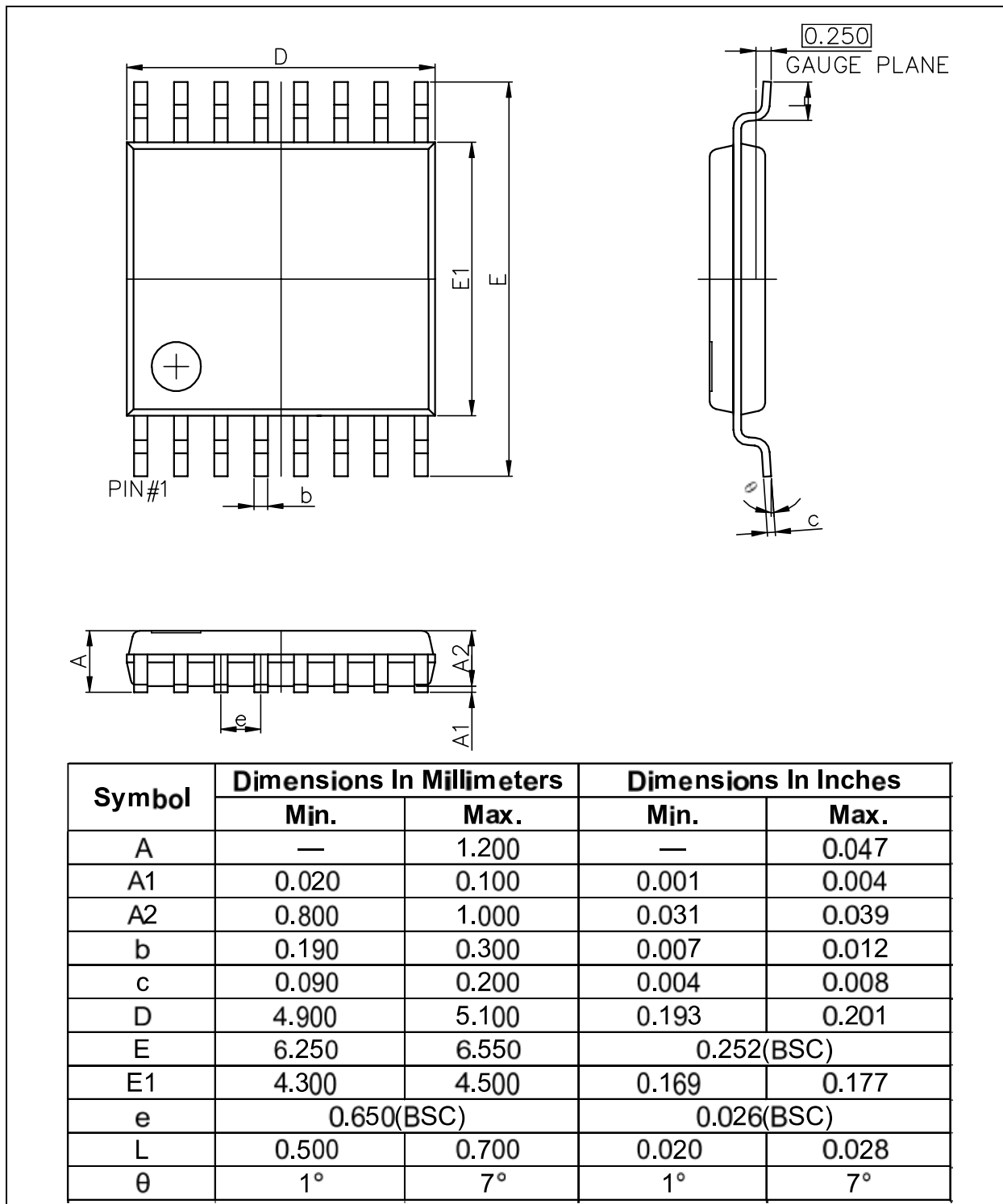


Symbol	Dimensions In Millimeters		Dimensions In Inches	
	Min.	Max.	Min.	Max.
A	—	1.750	—	0.069
A1	0.150	0.250	0.006	0.010
A2	1.400	1.500	0.055	0.059
b	0.330	0.510	0.013	0.020
c	0.170	0.250	0.007	0.010
D	9.800	10.000	0.386	0.394
e	1.270(BSC)		0.050(BSC)	
E	5.900	6.100	0.232	0.240
E1	3.800	4.000	0.150	0.157
L	0.400	1.270	0.016	0.050
θ	0°	8°	0°	8°

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TSSOP-16L



12. Abbreviations

Table 10. Abbreviations

Acronym	Description
CMOS	Complementary Metal-Oxide Semiconductor
DUT	Device Under Test
ESD	ElectroStatic Discharge
HBM	Human Body Model
CDM	Charged Device Model
TTL	Transistor-Transistor Logic

13. Revision History

Table 11. Revision history

Document ID	Release Date	Data sheet status	Change notice	Supersedes
EM74HC_HCT259 Rev. 1.0	Apr 28, 2025	Product datasheet		